Pontificia Universidad Catolica de Chile
Escuela de Ingeniería
Departamento de Ingeniería Mecánica y Metalúrgica
ICE3031 – Mecánica del Continuo

Tarea 5

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Tiempo dedicado: 21 hrs

Problema 1

Utilice el software `FEniCS` para resolver el problema de elasticidad no–lineal de un prisma sometido a una carga axial. Para esto, considere un prisma compuesto por un material Mooney–Rivlin compresible, cuya energía de deformación es

$$W = C_{10}\big(\bar{I}_1 - 3\big) + C_{01}\big(\bar{I}_2 - 3\big) + \frac{1}{D_1}(J-1)^2 \tag{1}$$

donde C_{10}, C_{01} y D_1 son parámetros del material,

$$\bar{I}_1 = J^{-2/3} I_1, \quad \bar{I}_2 = J^{-4/3} I_2$$

con I_1, I_2 los invariantes del tensor derecho de Cauchy–Green y J el campo jacobiano. El dominio a analizar es $B_0 = [0,4] \times [0,2] \times [0,2]$, cuya frontera es particionada como

$$\begin{aligned} \Gamma_{D_0} &= 0 \times (0,2) \times (0,2) \\ \Gamma_{D_1} &= 4 \times (0,2) \times (0,2) \\ \Gamma_N \ &= \partial\Omega \setminus \Gamma_D \end{aligned}$$

Para el problema, utilice las siguientes condiciones de frontera:

- En Γ_{D_1}:

$$u = (\alpha \cdot \Delta_{\max}, 0, 0)$$

- En Γ_{D_0}:

$$u = (0,0,0)$$

- En Γ_N:

$$T = (0,0,0)$$

Considerando $C_{10} = 0.8$, $C_{01} = 0.5$, $D_1 = 0.34$, $\Delta_{\max} = 4$ y ausencia de fuerzas de cuerpo, se pide

i) Escriba una implementación del problema usando el **principio de trabajos virtuales** en `FEniCS`, y resuélvalo. Utilizando $\alpha = 1$, entregue un gráfico donde se muestre la configuración de referencia B_0 y la configuración deformada $\varphi(B_0)$. Además, entregue un gráfico mostrando la magnitud del desplazamiento en el plano XZ.

ii) Para $\alpha = 1$, entregue gráficos 3D mostrando los planos XZ e XY para el jacobiano J y para cada una de las tensiones: $\sigma_{\mathrm{XX}}, \sigma_{\mathrm{YY}}, \sigma_{\mathrm{ZZ}}, \tau_{\mathrm{XY}}, \tau_{\mathrm{XZ}}, \tau_{\mathrm{YZ}}$.

iii) Entregue una animación (formato `.avi`) que muestre cómo se deforma el sólido a medida que se incrementa el desplazamiento. Para esto, resuelva el problema con distintos valores de $\alpha \in [0,1]$ (mínimo 10 valores distintos), y genere la animación utilizando `Paraview`.

iv) Entregue una curva fuerza–deformación.

Respuesta

i) Respuesta

Se utilizó el siguiente codigo en `Python` con la librería `FEniCSx 0.9.0` para implementar el principio de trabajos virtuales:

```
import numpy as np
import pyvista as pv
import ufl
from dolfinx import default_scalar_type, fem, mesh, plot
from dolfinx.fem.petsc import NonlinearProblem
from dolfinx.nls.petsc import NewtonSolver
from mpi4py import MPI

# Params
Lx = 4
Ly = 2
Lz = 2
nx = 40
ny = 20
nz = 20
tdim = 3

C10 = 0.8
C01 = 0.5
D1 = 0.34
D_max = 4
alpha = np.linspace(0, 1, 20+1)

comm = MPI.COMM_WORLD
cell_type = mesh.CellType.tetrahedron

domain = mesh.create_box(
    comm,
    [[0.0, 0.0, 0.0], [Lx, Ly, Lz]],
    [nx, ny, nz],
    cell_type=cell_type
)

C10 = fem.Constant(domain, default_scalar_type(C10))
C01 = fem.Constant(domain, default_scalar_type(C01))
D1 = fem.Constant(domain, default_scalar_type(D1))

B = fem.Constant(domain, default_scalar_type((0.0, 0.0, 0.0)))
T = fem.Constant(domain, default_scalar_type((0.0, 0.0, 0.0)))

V_space = fem.functionspace(domain, ("Lagrange", 1, (tdim,)))

# BC
def left_boundary(x): # Gamma_D0
    return np.isclose(x[0], 0.0)

def right_boundary(x): # Gamma_D1
    return np.isclose(x[0], Lx)

def every_other_boundary(x): # Gamma_N
    return np.isclose(x[1], 0.0) | np.isclose(x[1], Ly) | np.isclose(x[2],
0.0) | np.isclose(x[2], Lz)

left_dofs = fem.locate_dofs_geometrical(V_space, left_boundary)
right_dofs = fem.locate_dofs_geometrical(V_space, right_boundary)
bc1 = fem.dirichletbc(np.array([0.0, 0.0, 0.0], dtype=default_scalar_type),
left_dofs, V_space)
#lo definire en cada paso mejor
bc2 = fem.dirichletbc(np.array([alpha[-1]*D_max, 0.0, 0.0],
dtype=default_scalar_type), right_dofs, V_space)

other_facets = mesh.locate_entities_boundary(domain, tdim-1,
every_other_boundary)
facet_tags = mesh.meshtags(domain, tdim-1, other_facets,
np.full(other_facets.shape, 1, dtype=np.int32))
ds = ufl.Measure("ds", domain, subdomain_data=facet_tags)

def P_def(u, params: tuple, tdim=3):
    C10, C01, D1 = params
    I = ufl.Identity(tdim)
    F = ufl.variable(I + ufl.grad(u))
    J = ufl.det(F)
    C = ufl.dot(F.T, F)
    I1 = ufl.tr(C)
    I2 = 0.5 * (I1**2 - ufl.tr(ufl.dot(C, C)))

    I1bar = J**(-2/3) * I1
    I2bar = J**(-4/3) * I2

    W = C10 * (I1bar - 3) + C01 * (I2bar - 3) + 1/D1 * (J - 1)**2
    P = ufl.diff(W, F)
    return P

def residual(u, v, P, B, T):
    return ufl.inner(P(u), ufl.grad(v))*ufl.dx - ufl.dot(B,v)*ufl.dx -
ufl.dot(T,v)*ds(1)

def P(u): return P_def(u, (C10, C01, D1), tdim=tdim)

# desde aqui puedo recorrer la simulacion
u = fem.Function(V_space)
du = ufl.TrialFunction(V_space)
v = ufl.TestFunction(V_space)

R = residual(u, v, P, B, T)
DR = ufl.derivative(R, u, du)

bcs = [bc1, bc2]
problem = NonlinearProblem(R, u, bcs=bcs, J=DR)

solver = NewtonSolver(comm, problem)
solver.rtol = 1e-7
solver.atol = 1e-9
solver.max_it = 30
solver.report = True
ksp = solver.krylov_solver
ksp.setType("preonly")
pc = ksp.getPC()
pc.setType("lu")
pc.setFactorSolverType("mumps")

I = ufl.Identity(tdim)
F = I + ufl.grad(u)
J = ufl.det(F)
_P = P(u)
sigma = ufl.dot(_P, F.T) / J

V_J = fem.functionspace(domain, ("DG", 0))
V_sigma = fem.functionspace(domain, ("DG", 0, (tdim, tdim)))

expr_J = fem.Expression(J, V_J.element.interpolation_points())
expr_sigma = fem.Expression(sigma, V_sigma.element.interpolation_points())

uhs = []
js = []
sigmas = []

for a in alpha:
    print(f"alpha = {a:.2f}", end=", ")
    bc2 = fem.dirichletbc(np.array([a*D_max, 0.0, 0.0],
dtype=default_scalar_type), right_dofs, V_space)
    problem.bcs = [bc1, bc2]

    nit, flag = solver.solve(u)
    u.x.scatter_forward()
    print(f"Number of Newton iterations: {nit}, Converged: {flag}")

    u_copy = fem.Function(V_space)
    J_func = fem.Function(V_J, name="J")
    sigma_func = fem.Function(V_sigma, name="Sigma")

    u_copy.x.array[:] = u.x.array[:]
    J_func.interpolate(expr_J)
    sigma_func.interpolate(expr_sigma)
    J_func.x.scatter_forward()
    sigma_func.x.scatter_forward()

    uhs.append(u_copy)
    js.append(J_func)
    sigmas.append(sigma_func)
```

Para las simulaciones se utilizó un tamaño de malla de `(nx, ny, nz) = (40, 20, 20)`, y para la animación se utilizaron los siguientes valores de α: `alpha = np.linspace(0, 1, 21)`. Con esto, en la Figura 1 se muestra en negro el prisma en la configuración de referencia B_0 y en azul la configuración actual $\boldsymbol{\varphi}(B_0)$, para $\alpha = 1$.

En la Figura 2 se muestra la magnitud del desplazamiento para el plano X–Z en la configuración deformada $\boldsymbol{\varphi}(B_0)$. Se puede apreciar que la magnitud del extremo derecho es equivalente al desplazamiento prescrito en Γ_{D_1} para $\alpha = 1$, $\Gamma_{D_1}(\alpha = 1) = (4, 0, 0)$.

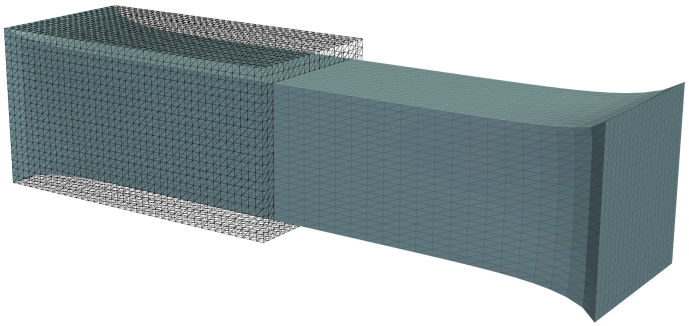


Figura 1: Simulación del prisma en carga axial ($\alpha = 1$), con B_0 en negro y $\boldsymbol{\varphi}(B_0)$ en azul.

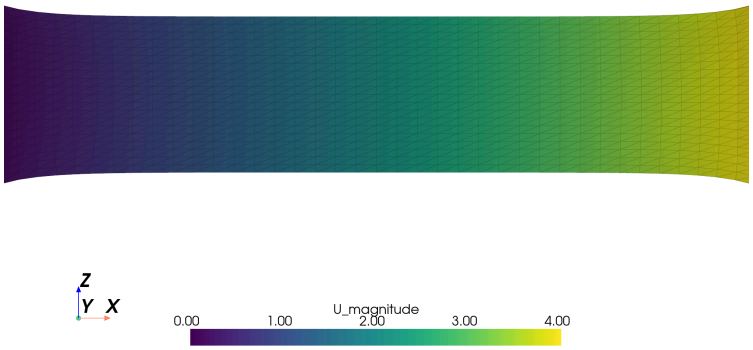


Figura 2: Magnitud del desplazamiento en el plano X–Z en $\boldsymbol{\varphi}(B_0)$

ii) Respuesta

Las Figuras 3 – 8 muestran los campos de tensión $\sigma_{XX}, \sigma_{YY}, \sigma_{ZZ}, \tau_{XY}, \tau_{XZ}, \tau_{YZ}$ en los planos X–Y, X–Z en la configuración deformada $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$, mientras que la Figura 9 muestra el campo jacobiano de la misma forma.

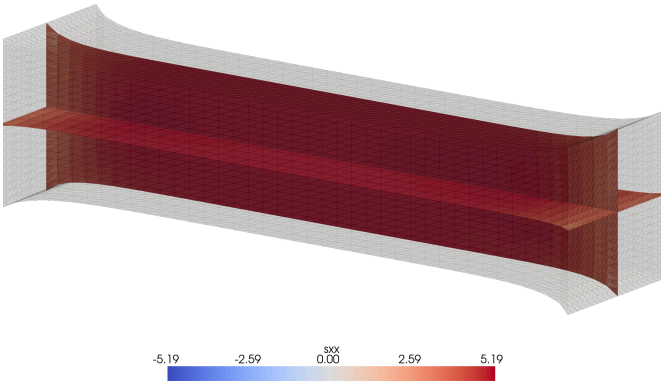


Figura 3: campo σ_{xx} en $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$

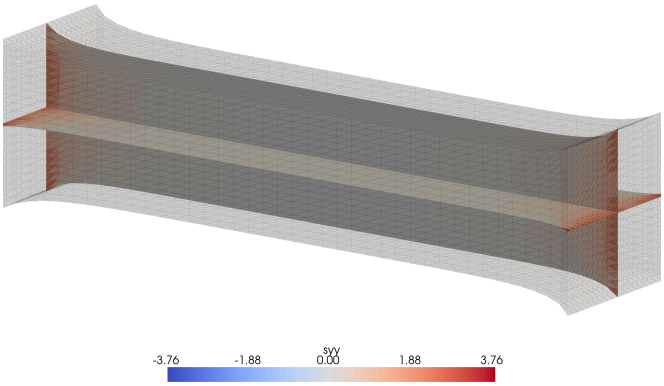


Figura 4: campo σ_{yy} en $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$

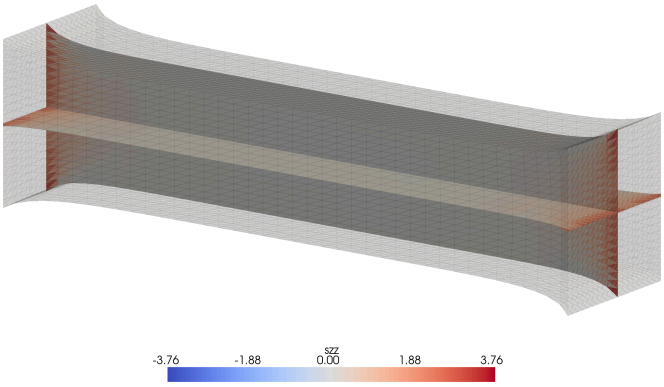


Figura 5: campo σ_{zz} en $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$

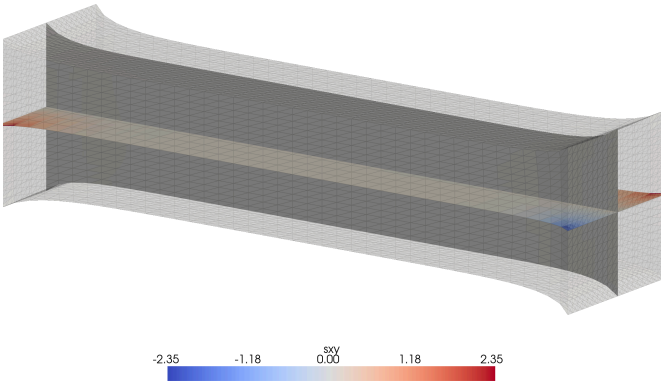


Figura 6: campo τ_{xy} en $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$

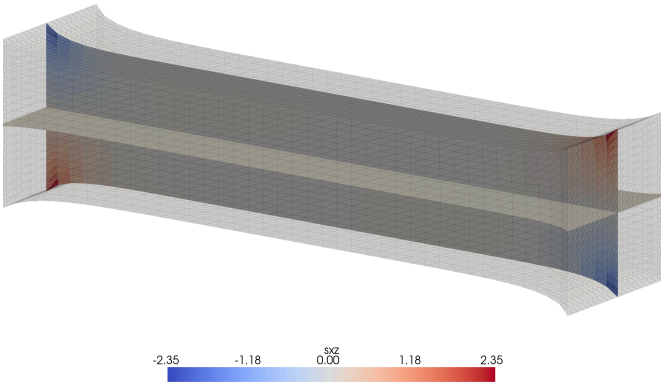


Figura 7: campo τ_{xz} en $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$

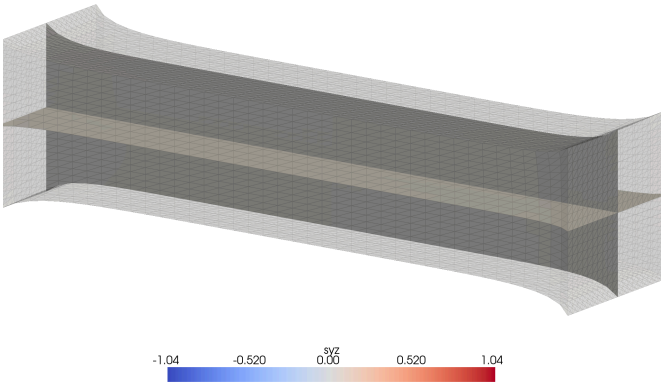


Figura 8: campo τ_{yz} en $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$

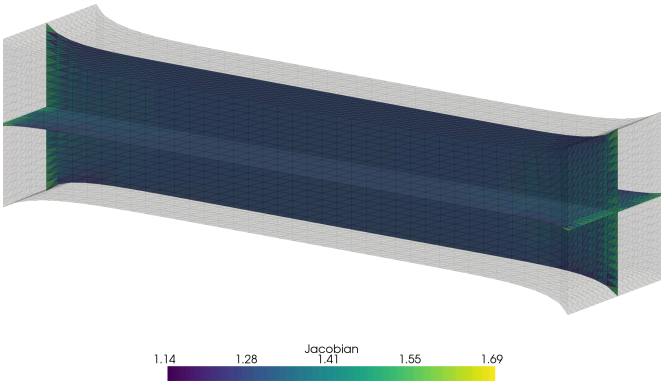


Figura 9: Campo J en $\boldsymbol{\varphi}(B_0)$ para $\alpha = 1$

iii) Respuesta

La Figura 10 muestra la vista previa de la animación en formato `.avi` adjunto a la entrega.

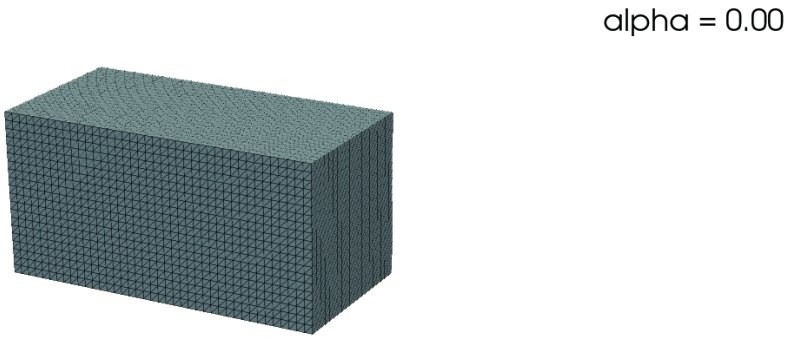


Figura 10: Animación del desplazamiento del primsa para $\alpha \in [0, 1]$ (21 frames).

Problema 2

i) Considere un material isotrópico cuya energía de deformación es $W(\boldsymbol{C}) = \widetilde{W}(I_1(\boldsymbol{C}), I_2(\boldsymbol{C}), I_3(\boldsymbol{C}))$. Demuestre que para este caso es posible escribir el segundo tensor de Piola–Kirchhoff como

$$\boldsymbol{S} = \alpha_0(I_1, I_2, I_3)\boldsymbol{I} + \alpha_1(I_1, I_2, I_3)\boldsymbol{C} + \alpha_2(I_1, I_2, I_3)\boldsymbol{C}^2 \tag{2}$$

donde

$$\begin{aligned}
\alpha_0(I_1, I_2, I_3) &= 2\frac{\partial W}{\partial I_1} + 2\frac{\partial W}{\partial I_2}I_1 + 2\frac{\partial W}{\partial I_3}I_2 \\
\alpha_1(I_1, I_2, I_3) &= -2\frac{\partial W}{\partial I_2} - 2\frac{\partial W}{\partial I_3}I_1 \\
\alpha_2(I_1, I_2, I_3) &= 2\frac{\partial W}{\partial I_3}
\end{aligned}$$

ii) Considere un sólido incompresible en configuración de referencia B_0 cuyo Lagrangiano es definido como

$$I[\boldsymbol{\varphi}, p] \coloneqq \Pi[\boldsymbol{\varphi}] - \int_{B_0} p\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V \tag{3}$$

y donde las configuraciones de equilibrio se obtienen de la condición de estacionariedad

$$\delta I[\boldsymbol{\varphi}, p, \boldsymbol{\eta}, \xi] \coloneqq \frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi} + \epsilon\boldsymbol{\eta}, p + \epsilon\xi]\Big|_{\epsilon=0} = 0 \tag{4}$$

Demuestre que lo anaterior es equivalente a

$$\begin{aligned}
\boldsymbol{\nabla}_0 \cdot \bar{\boldsymbol{P}} + R\boldsymbol{B} = 0 \qquad & \boldsymbol{X} \in B_0 \\
\bar{\boldsymbol{P}}\widehat{\boldsymbol{N}} = \bar{\boldsymbol{T}} \qquad & \boldsymbol{X} \in \partial B_2 \\
\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1 = 0 \qquad & \boldsymbol{X} \in B_0
\end{aligned}$$

donde

$$\bar{\boldsymbol{P}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}, p) = \frac{\partial W}{\partial \boldsymbol{F}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - p\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})\{\boldsymbol{\nabla}_0\boldsymbol{\varphi}\}^{-\top} \tag{5}$$

y ∂B_2 es la frontera de tracciones prescritas.

i) Respuesta

Del teorema de Coleman–Noll, para una función de energía de deformación W, el segundo tensor de Piola–Kirchhoff es

$$\boldsymbol{S} = 2\frac{\partial W}{\partial \boldsymbol{C}} \tag{6}$$

Entonces, para una funcion de energia $W(C) = \widetilde{W}(I_1(\boldsymbol{C}), I_2(\boldsymbol{C}), I_3(\boldsymbol{C}))$ se aplica la regla de la cadena como

$$\boldsymbol{S} = 2\frac{\partial W}{\partial \boldsymbol{C}} = 2\frac{\partial \widetilde{W}}{\partial I_1}\frac{\partial I_1}{\partial \boldsymbol{C}} + 2\frac{\partial \widetilde{W}}{\partial I_2}\frac{\partial I_2}{\partial \boldsymbol{C}} + 2\frac{\partial \widetilde{W}}{\partial I_3}\frac{\partial I_3}{\partial \boldsymbol{C}} \tag{7}$$

donde las invariantes $I_1(\boldsymbol{C})$, $I_1(\boldsymbol{C})$ y $I_1(\boldsymbol{C})$ son

$$\begin{aligned} I_1(\boldsymbol{C}) &= \operatorname{tr}(\boldsymbol{C}) \\ I_2(\boldsymbol{C}) &= \frac{1}{2}(\operatorname{tr}^2(\boldsymbol{C}) - \operatorname{tr}(\boldsymbol{C}^2)) \\ I_3(\boldsymbol{C}) &= \det(\boldsymbol{C}) \end{aligned}$$

entonces se tendrá (y usando que $\boldsymbol{C} = \boldsymbol{C}^\top$)

$$\begin{aligned} \frac{\partial I_1}{\partial \boldsymbol{C}} &= \frac{\partial \operatorname{tr}(\boldsymbol{C})}{\partial \boldsymbol{C}} \equiv \frac{\partial C_{KK}}{\partial C_{IJ}} = \delta_{KI}\delta_{KJ} = \delta_{IJ} \equiv \boldsymbol{I} \\ \frac{\partial \operatorname{tr}^2(\boldsymbol{C})}{\partial \boldsymbol{C}} &\equiv \frac{\partial C_{KK}C_{LL}}{\partial C_{IJ}} = 2C_{KK}\delta_{IJ} \equiv 2\operatorname{tr}(\boldsymbol{C})\boldsymbol{I} = 2I_1\boldsymbol{I} \\ \frac{\partial \operatorname{tr}(\boldsymbol{C}^2)}{\partial \boldsymbol{C}} &\equiv \frac{\partial C_{KL}C_{LK}}{\partial C_{IJ}} = C_{LK}\delta_{IK}\delta_{JL} + C_{KL}\delta_{IL}\delta_{JK} = 2C_{JI} \equiv 2\boldsymbol{C} \\ \frac{\partial I_2}{\partial \boldsymbol{C}} &= \frac{1}{2}\left(\frac{\partial \operatorname{tr}^2(\boldsymbol{C})}{\partial \boldsymbol{C}} - \frac{\partial \operatorname{tr}(\boldsymbol{C}^2)}{\partial \boldsymbol{C}}\right) = I_1\boldsymbol{I} - \boldsymbol{C} \\ \frac{\partial I_3}{\partial \boldsymbol{C}} &= \frac{\partial \det(\boldsymbol{C})}{\partial \boldsymbol{C}} := \det(\boldsymbol{C})\boldsymbol{C}^{-\top} = I_3\boldsymbol{C}^{-1} \end{aligned}$$

Como no nos sirve esa expresión para $\frac{\partial I_3}{\partial \boldsymbol{C}}$, hacemos lo siguiente

$$\begin{aligned} \frac{\partial \det(\boldsymbol{C})}{\partial \boldsymbol{C}} &\equiv \frac{\partial}{\partial C_{IJ}}\left(\frac{1}{6}\epsilon_{abc}\epsilon_{ijk}C_{ai}C_{bj}C_{ck}\right) \\ &= \frac{1}{6}\epsilon_{abc}\epsilon_{ijk}\left(\delta_{Ia}\delta_{Ji}C_{bj}C_{ck} + C_{ai}\delta_{Ib}\delta_{Jj}C_{ck} + C_{ai}C_{bj}\delta_{Ic}\delta_{Jk}\right) \\ &= \frac{1}{6}\left(\epsilon_{abc}\epsilon_{ijk}\delta_{Ia}\delta_{Ji}C_{bj}C_{ck} + \epsilon_{abc}\epsilon_{ijk}C_{ai}\delta_{Ib}\delta_{Jj}C_{ck} + \epsilon_{abc}\epsilon_{ijk}C_{ai}C_{bj}\delta_{Ic}\delta_{Jk}\right) \\ &= \frac{1}{6}(\epsilon_{Ibc}\epsilon_{Jjk}C_{bj}C_{ck} + \epsilon_{aIc}\epsilon_{iJk}C_{ai}C_{ck} + \epsilon_{abI}\epsilon_{ijJ}C_{ai}C_{bj} \end{aligned}$$

$$= \frac{1}{2}\epsilon_{IAM}\epsilon_{JBN}C_{AB}C_{MN}$$

$$= \frac{1}{2}(\delta_{IJ}(\delta_{AB}\delta_{MN} - \delta_{AN}\delta_{MB}) - \delta_{IB}(\delta_{AJ}\delta_{MN} - \delta_{AN}\delta_{MJ}) + \delta_{IN}(\delta_{AJ}\delta_{MB} - \delta_{AB}\delta_{MJ}))C_{AB}C_{MN}$$

$$= \frac{1}{2}(\delta_{IJ}(C_{AA}C_{MM} - C_{AB}C_{BA}) - (C_{JI}C_{MM} - C_{AI}C_{JA}) + (C_{JB}C_{BI} - C_{AA}C_{JI}))$$

$$= \frac{1}{2}\big(\delta_{IJ}\big(C_{AA}^2 - C_{AB}C_{BA}\big) - 2C_{JI}C_{AA} + 2C_{JA}C_{AI}\big)$$

$$= \frac{1}{2}\delta_{IJ}\big(C_{AA}^2 - C_{AB}C_{BA}\big) - C_{JI}C_{AA} + C_{JA}C_{AI}$$

$$= \frac{1}{2}(\mathrm{tr}^2(C) - \mathrm{tr}(\boldsymbol{C}\boldsymbol{C}))\boldsymbol{I} - \boldsymbol{C}^\top \,\mathrm{tr}\,\boldsymbol{C} + \boldsymbol{C}^\top\boldsymbol{C}^\top$$

$$\frac{\partial \det(\boldsymbol{C})}{\partial \boldsymbol{C}} = I_2\boldsymbol{I} - I_1\boldsymbol{C} + \boldsymbol{C}^2$$

Con esto, se reescribe (7) como

$$\boldsymbol{S} = 2\frac{\partial \widetilde{W}}{\partial I_1}\boldsymbol{I} + 2\frac{\partial \widetilde{W}}{\partial I_2}(I_1\boldsymbol{I} - \boldsymbol{C}) + 2\frac{\partial \widetilde{W}}{\partial I_3}(I_2\boldsymbol{I} - I_1\boldsymbol{C} + \boldsymbol{C}^2)$$

$$\boldsymbol{S} = \underbrace{\left(2\frac{\partial \widetilde{W}}{\partial I_1} + 2\frac{\partial \widetilde{W}}{\partial I_2}I_1 + 2\frac{\partial \widetilde{W}}{\partial I_3}I_2\right)}_{\alpha_0(I_1,I_2,I_3)}\boldsymbol{I} + \underbrace{\left(-2\frac{\partial \widetilde{W}}{\partial I_2} - 2\frac{\partial \widetilde{W}}{\partial I_3}I_1\right)}_{\alpha_1(I_1,I_2,I_3)}\boldsymbol{C} + \underbrace{\left(2\frac{\partial \widetilde{W}}{\partial I_3}\right)}_{\alpha_2(I_1,I_2,I_3)}\boldsymbol{C}^2$$

$$\boldsymbol{S} = \alpha_0(I_1, I_2, I_3)\boldsymbol{I} + \alpha_1(I_1, I_2, I_3)\boldsymbol{C} + \alpha_2(I_1, I_2, I_3)\boldsymbol{C}^2$$

■

ii) Respuesta

Sea $\Pi[\boldsymbol{\varphi}]$ la energía potencial total, según [1], como la enegía de deformación del cuerpo menos el trabajo por las cargas (potenciales) aplicadas, como

$$\Pi[\boldsymbol{\varphi}] = \int_{B_0} W(\boldsymbol{F})\,\mathrm{d}V - \int_{B_0} R\boldsymbol{B}\cdot\boldsymbol{\varphi}\,\mathrm{d}V - \int_{\partial B_0} \boldsymbol{T}\cdot\boldsymbol{\varphi}\,\mathrm{d}S \tag{8}$$

con esto, se tiene el siguiente Lagrangiano

$$I[\boldsymbol{\varphi}, p] = \int_{B_0} W(\boldsymbol{F})\,\mathrm{d}V - \int_{B_0} R\boldsymbol{B}\cdot\boldsymbol{\varphi}\,\mathrm{d}V - \int_{\partial B_0} \boldsymbol{T}\cdot\boldsymbol{\varphi}\,\mathrm{d}S - \int_{B_0} p\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V$$

Las configuraciones de equilibrio estan dados por la condición de estacionariedad, que se consigue minimizando el Lagrangiano para una variación arbitraria $(\delta p, \delta\boldsymbol{\varphi}) = (\xi, \boldsymbol{\eta})$

$$\delta I[\boldsymbol{\varphi}, p, \boldsymbol{\eta}, \xi] := \frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi} + \epsilon\boldsymbol{\eta}, p + \epsilon\xi]\Big|_{\epsilon=0} = 0$$

desarrollando por cada término, se tiene

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}, p+\epsilon\xi]\Big|_{\epsilon=0} &= \frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(\int_{B_0} W(\boldsymbol{\nabla}_0(\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}))\,\mathrm{d}V\right)\Big|_{\epsilon=0} + \\
&\quad -\frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(\int_{B_0} R\boldsymbol{B}\cdot(\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta})\,\mathrm{d}V\right)\Big|_{\epsilon=0} + \\
&\quad -\frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(\int_{\partial B_0} \boldsymbol{T}\cdot(\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta})\,\mathrm{d}S\right)\Big|_{\epsilon=0} + \\
&\quad -\frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(\int_{B_0} (p+\epsilon\xi)\{\det(\boldsymbol{\nabla}_0(\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}))-1\}\,\mathrm{d}V\right)\Big|_{\epsilon=0} \\
\frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}, p+\epsilon\xi]\Big|_{\epsilon=0} &= \int_{B_0} \frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(W(\boldsymbol{\nabla}_0(\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}))\right)\Big|_{\epsilon=0}\mathrm{d}V + \\
&\quad -\int_{B_0} \frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(R\boldsymbol{B}\cdot\boldsymbol{\varphi}+\epsilon R\boldsymbol{B}\cdot\boldsymbol{\eta}\right)\Big|_{\epsilon=0}\mathrm{d}V + \\
&\quad -\int_{\partial B_0} \frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(\boldsymbol{T}\cdot\boldsymbol{\varphi}+\epsilon\boldsymbol{T}\cdot\boldsymbol{\eta}\right)\Big|_{\epsilon=0}\mathrm{d}S + \\
&\quad -\int_{B_0} \frac{\mathrm{d}}{\mathrm{d}\epsilon}\left((p+\epsilon\xi)\{\det(\boldsymbol{\nabla}_0(\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}))-1\}\right)\Big|_{\epsilon=0}\mathrm{d}V \\
\frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}, p+\epsilon\xi]\Big|_{\epsilon=0} &= \int_{B_0} \frac{\partial W}{\partial \boldsymbol{F}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) : \frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(\boldsymbol{\nabla}_0\boldsymbol{\varphi}+\epsilon\boldsymbol{\nabla}_0\boldsymbol{\eta}\right)\Big|_{\epsilon=0}\mathrm{d}V + \\
&\quad -\int_{B_0} R\boldsymbol{B}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{\partial B_0}\boldsymbol{T}\cdot\boldsymbol{\eta}\,\mathrm{d}S + \\
&\quad -\int_{B_0} p\frac{\mathrm{d}}{\mathrm{d}\epsilon}\left(\det(\boldsymbol{\nabla}_0(\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}))\right)\Big|_{\epsilon=0} + \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})-1\}\,\mathrm{d}V \\
\frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}, p+\epsilon\xi]\Big|_{\epsilon=0} &= \int_{B_0} \frac{\partial W}{\partial \boldsymbol{F}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) : (\boldsymbol{\nabla}_0\boldsymbol{\eta})\,\mathrm{d}V + \\
&\quad -\int_{B_0} R\boldsymbol{B}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{\partial B_0}\boldsymbol{T}\cdot\boldsymbol{\eta}\,\mathrm{d}S + \\
&\quad -\int_{B_0} p\underbrace{\mathcal{D}\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})\}\cdot\boldsymbol{\eta}}_{\det(\boldsymbol{F})(\boldsymbol{F}^{-\top}:\boldsymbol{\nabla}_0\boldsymbol{\eta})} + \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})-1\}\,\mathrm{d}V \\
\frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}, p+\epsilon\xi]\Big|_{\epsilon=0} &= \int_{B_0} \frac{\partial W}{\partial \boldsymbol{F}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) : \boldsymbol{\nabla}_0\boldsymbol{\eta}\,\mathrm{d}V - \int_{B_0} R\boldsymbol{B}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{\partial B_0}\boldsymbol{T}\cdot\boldsymbol{\eta}\,\mathrm{d}S + \\
&\quad -\int_{B_0} p\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})\{\boldsymbol{\nabla}_0\boldsymbol{\varphi}\}^{-\top} : \boldsymbol{\nabla}_0\boldsymbol{\eta} + \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})-1\}\,\mathrm{d}V
\end{aligned}$$

$$\left.\frac{\mathrm{d}}{\mathrm{d}\epsilon} I[\boldsymbol{\varphi}+\epsilon\boldsymbol{\eta}, p+\epsilon\xi]\right|_{\epsilon=0} = \int_{B_0} \underbrace{\left(\frac{\partial W}{\partial \boldsymbol{F}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - p\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})\{\boldsymbol{\nabla}_0\boldsymbol{\varphi}\}^{-\top}\right)}_{\bar{\boldsymbol{P}}} : \boldsymbol{\nabla}_0\boldsymbol{\eta} - R\boldsymbol{B}\cdot\boldsymbol{\eta}\,\mathrm{d}V$$

$$- \int_{\partial B_0} \boldsymbol{T}\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V$$

$$\delta I[\boldsymbol{\varphi}, p, \boldsymbol{\eta}, \xi] = \int_{B_0} \underbrace{\bar{\boldsymbol{P}} : \boldsymbol{\nabla}_0\boldsymbol{\eta}}_{\boldsymbol{\nabla}_0\cdot(\bar{\boldsymbol{P}}^\top\boldsymbol{\eta}) - (\boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}})\cdot\boldsymbol{\eta}} - R\boldsymbol{B}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{\partial B_0} \boldsymbol{T}\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V$$

$$\delta I[\boldsymbol{\varphi}, p, \boldsymbol{\eta}, \xi] = \int_{B_0} \boldsymbol{\nabla}_0\cdot\left(\bar{\boldsymbol{P}}^\top\boldsymbol{\eta}\right)\mathrm{d}V - \int_{B_0} \left(\boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}}\right)\cdot\boldsymbol{\eta} + R\boldsymbol{B}\cdot\boldsymbol{\eta}\,\mathrm{d}V +$$

$$- \int_{\partial B_0} \boldsymbol{T}\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V$$

$$\delta I[\boldsymbol{\varphi}, p, \boldsymbol{\eta}, \xi] = \int_{\partial B_0} \underbrace{\left(\bar{\boldsymbol{P}}^\top\boldsymbol{\eta}\right)\cdot\widehat{\boldsymbol{N}}}_{(\bar{\boldsymbol{P}}\widehat{\boldsymbol{N}})\cdot\boldsymbol{\eta}}\,\mathrm{d}S - \int_{B_0} \left(\boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}} + R\boldsymbol{B}\right)\cdot\boldsymbol{\eta}\,\mathrm{d}V +$$

$$- \int_{\partial B_0} \boldsymbol{T}\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V$$

$$\delta I[\boldsymbol{\varphi}, p, \boldsymbol{\eta}, \xi] = \int_{\partial B_0} \left(\bar{\boldsymbol{P}}\widehat{\boldsymbol{N}} - \boldsymbol{T}\right)\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \left(\boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}} + R\boldsymbol{B}\right)\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{B_0} \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V$$

Separando la frontera $\partial B_0 = \partial B_1 \cup \partial B_2$, con $\boldsymbol{T} = 0$ en ∂B_1, y $\boldsymbol{T} = \bar{\boldsymbol{T}}$ en ∂B_2

$$\delta I[\boldsymbol{\varphi}, p, \boldsymbol{\eta}, \xi] = \int_{\partial B_2} \left(\bar{\boldsymbol{P}}\widehat{\boldsymbol{N}} - \bar{\boldsymbol{T}}\right)\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \left(\boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}} + R\boldsymbol{B}\right)\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{B_0} \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V$$

$$\int_{\partial B_2} \left(\bar{\boldsymbol{P}}\widehat{\boldsymbol{N}} - \bar{\boldsymbol{T}}\right)\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \left(\boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}} + R\boldsymbol{B}\right)\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{B_0} \xi\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}\,\mathrm{d}V = 0, \quad \forall(\xi, \boldsymbol{\eta})$$

Como se debe cumplir para $(\xi, \boldsymbol{\eta})$ arbitrarios, entonces necesariamente cada término debe ser 0

$$\int_{\partial B_2} \underbrace{\left(\bar{\boldsymbol{P}}\widehat{\boldsymbol{N}} - \bar{\boldsymbol{T}}\right)}_{\equiv\boldsymbol{0}}\cdot\boldsymbol{\eta}\,\mathrm{d}S - \int_{B_0} \underbrace{\left(\boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}} + R\boldsymbol{B}\right)}_{\equiv\boldsymbol{0}}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{B_0} \xi\underbrace{\{\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1\}}_{\equiv 0}\,\mathrm{d}V = 0, \quad \forall(\xi, \boldsymbol{\eta})$$

Finalmente, esto es equivalente a

$$\begin{aligned} \boldsymbol{\nabla}_0\cdot\bar{\boldsymbol{P}} + R\boldsymbol{B} &= 0, \quad \boldsymbol{X}\in B_0 \\ \bar{\boldsymbol{P}}\widehat{\boldsymbol{N}} &= \bar{\boldsymbol{T}}, \quad \boldsymbol{X}\in\partial B_2 \\ \det(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - 1 &= 0, \quad \boldsymbol{X}\in B_0 \end{aligned} \tag{9}$$

$$\bar{\boldsymbol{P}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}, p) = \frac{\partial W}{\partial \boldsymbol{F}}(\boldsymbol{\nabla}_0\boldsymbol{\varphi}) - p\det(\boldsymbol{\nabla}_0\boldsymbol{\varphi})\{\boldsymbol{\nabla}_0\boldsymbol{\varphi}\}^{-\top}$$

■

Uso de herramientas de IA

En esta tarea no se usaron herramientas de IA.

Bibliografía

[1] Ellad B. Tadmor, Ronald E. Miller, y Ryan S. Elliott, *Continuum Mechanics and Thermodynamics: From Fundamental Concepts to Governing Equations*. Cambridge University Press, 2011.